

FE-APP-03 - Contractor Technical Mobile App

Version: v1.0

Status: Draft for review

Depends on:

- FE-APP-00-Frontend_Architecture_and_Shared_UX_Foundation-v1.0
- DD_CROSS-00-System_Traceability_and_Implementation_Map-v1.0
- DD_API-00-API_Catalog_and_Gateway_Standards-v1.0
- DD_QA-01-End_to_End_Test_Strategy-v1.0
- DD_WO-01-Work_Order_Module-v1.0
- DD_WO-01-FRAMEWORK-v1.0
- DD_WO-01-FLOW-INSTALLATION-v1.0
- DD_WO-01-FLOW-SHIFTING-v1.0
- DD_WO-01-FLOW-SUPPORT-v1.0
- DD_OSR-01-Stock_Chain_Engine-v1.0
- DD_OSR-INSTANCE-01-Equipment_Instance_Registry-v1.0
- DD_OSR-RMA-01-Equipment_Swap_and_RMA-v1.0
- DD_EM-02-Contractor_and_Staff_Registry-v1.0
- DD_EM-02-Contractor_Team_Agent_Registry-v1.0
- DD_TCK-01-Ticketing_and_Case_Management-v1.0
- FOUNDATION_AUTH
- FOUNDATION_FILE_STORAGE
- DD_ICN-01-Internal_Communications-v1.0

1. Purpose

This DD defines the Contractor Technical Mobile App.

The app is used by field technicians, contractor team leads, and internal technical teams to receive assigned work orders, claim/start jobs, view customer/site context, capture checklist evidence, scan equipment serials, upload photos/signatures, submit finalization, and handle support/escalation messages.

The app is Kenya/WIK-first but must support other operators through configuration: language, brand, work order types, checklist rules, evidence requirements, technician roles, and offline policy.

2. Design Position

FE-APP-03 is the mobile execution surface for work orders. It is not the Work Order system of record.

Ownership split:

Concern	Owner
Work order lifecycle, assignment, claim, start, notes, attachments, finalization	WO-01
Contractor, team, technician identity and eligibility	EM-02
Equipment serial validation, stock, instance lifecycle	OSR-01, OSR-INSTANCE-01, OSR-RMA-01
Photos, signatures, documents, evidence files	FOUNDATION_FILE_STORAGE
Staff task notifications and operational messages	ICN-01
Ticket context for support WOs	TCK-01
Customer/contact/subscription context shown on WO	Read through WO-01 detail or owner APIs where authorized
Fulfillment/order context for installation WOs	FUL-02, usually surfaced through WO context

Illustration:

Figure 1 - Contractor app module map



3. Application Scope

Included:

Area	Included features	Primary module APIs
Job queue	Assigned jobs, team queue, filters by status/type/region.	WO-01, EM-02.
Job detail	Customer/site/contact/task/checklist/equipment context.	WO-01, read context from ILM/SUB/FUL through allowed APIs where needed.
Claim/start	Claim team job, start job, mark arrival where supported.	WO-01.
Checklist	Installation/support/shifting checklist capture.	WO-01 flow satellites.
Evidence capture	Photos, signatures, notes, attachments.	FOUNDATION_FILE_STORAGE, WO-01.
Equipment scan	Scan/validate serials, bind/recover devices, show technician stock.	OSR-INSTANCE-01, OSR-01, WO-01.
Finalization	Submit final result, reason codes, evidence, equipment bindings.	WO-01, OSR where equipment changes are required.

Area	Included features	Primary module APIs
Offline job workspace	Cache assigned WO detail, capture draft evidence, sync later.	Local encrypted store; server authority remains WO/OSR/file APIs.
Messages/tasks	Staff messages, supervisor requests, correction requests.	ICN-01, WO-01.
Support escalation	Capture issue and create/request escalation where allowed.	WO-01, TCK-01 if ticket lifecycle is needed.

Non-scope:

Non-scope	Owner
Dispatch planning and reassignment board	FE-APP-01, WO-01.
Ticket lifecycle management	TCK-01, mostly FE-APP-01/FE-APP-04.
Stock warehouse administration	FE-APP-01, OSR admin screens.
Customer self-service appointment changes	FE-APP-04 if enabled.
Contractor performance reporting	FE-APP-05, REP-01.

4. Users And Roles

Roles come from FOUNDATION_AUTH. Contractor/team/staff identity comes from EM-02.

Role family	Typical permissions	Module API link
Field technician	View assigned WOs, claim/start, capture evidence, submit finalization.	FOUNDATION_AUTH, EM-02, WO-01.
Contractor team lead	View team queue, assign/claim within team if allowed, review pending evidence.	EM-02, WO-01.
Contractor dispatcher	Manage contractor/team queue, reassign within allowed franchise/team.	EM-02, WO-01.
Internal technical team	Same execution functions, broader operator scope where authorized.	FOUNDATION_AUTH, EM-02, WO-01, OSR.
Support/QCS technician	Execute support or QCS WOs and capture escalation outcome.	WO-01, TCK-01 context where linked.
Read-only auditor	View assigned/completed work and evidence metadata where allowed.	WO-01, file metadata APIs.

Backend authorization remains authoritative. The app must not allow a user to mutate a WO simply because it was cached locally.

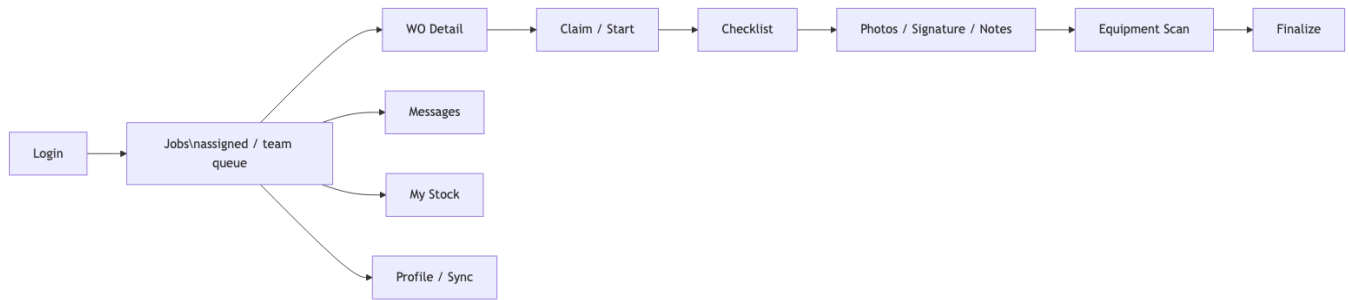
5. Navigation Model

Recommended mobile navigation:

Tab	Purpose	Module API link
Jobs	Assigned WOs, team queue, due/urgent work.	WO-01, EM-02.
Map	Site route/location view if enabled.	WO-01 location context, operator map config.
Stock	Technician/team stock and scanned equipment.	OSR-01, OSR-INSTANCE-01.
Messages	Staff messages, correction requests, supervisor notes.	ICN-01, WO-01.
Profile	User, team, contractor, sync status, logout.	FOUNDATION_AUTH, EM-02, runtime config.

Illustration:

Figure 2 - Contractor app screen flow



6. Application Shell

Required shell components:

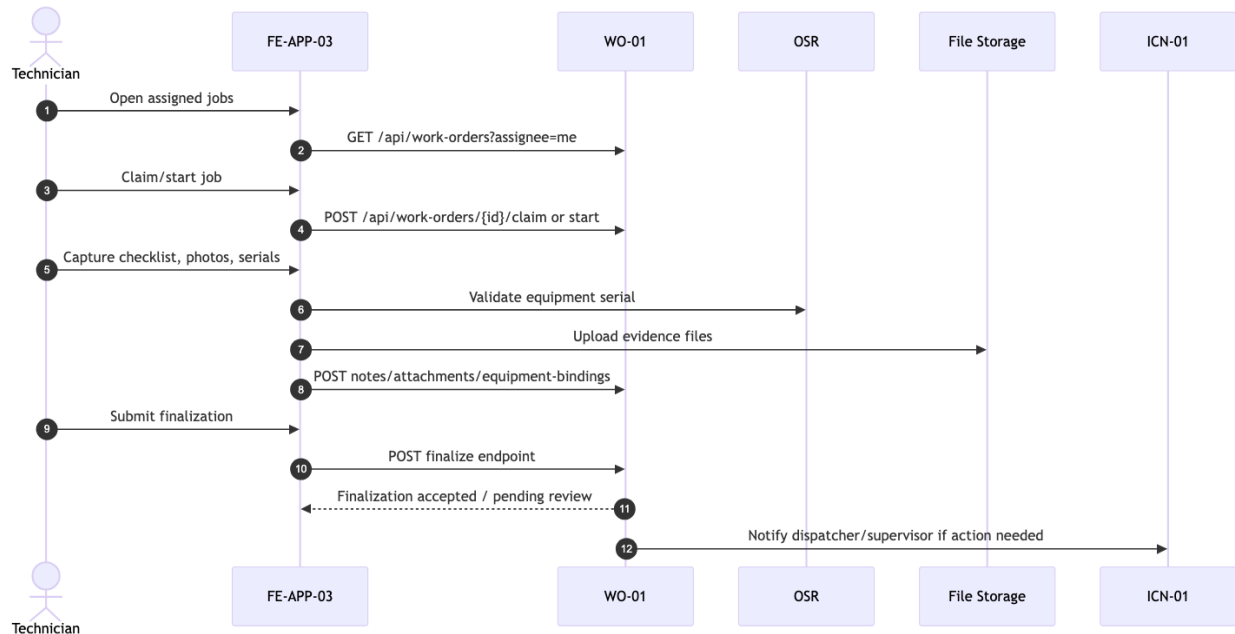
Component	Behavior	Module/config link
Operator context	Shows operator/country/brand.	Runtime config from FE-APP-00.
Offline indicator	Shows online/offline/sync pending.	Local sync manager.
Job sync badge	Shows unsynced evidence/finalization attempts.	Local queue plus WO/file/OSR API responses.
Bottom navigation	Jobs, Map, Stock, Messages, Profile.	Frontend route config.
Quick scan	Opens camera/serial scanner from job or stock.	OSR - INSTANCE - 01, WO - 01.
Profile	User, role, contractor/team, logout.	FOUNDATION_AUTH, EM-02.

7. Main User Flows

7.1 Work Order Execution

Illustration:

Figure 3 - Work order execution flow



Flow:

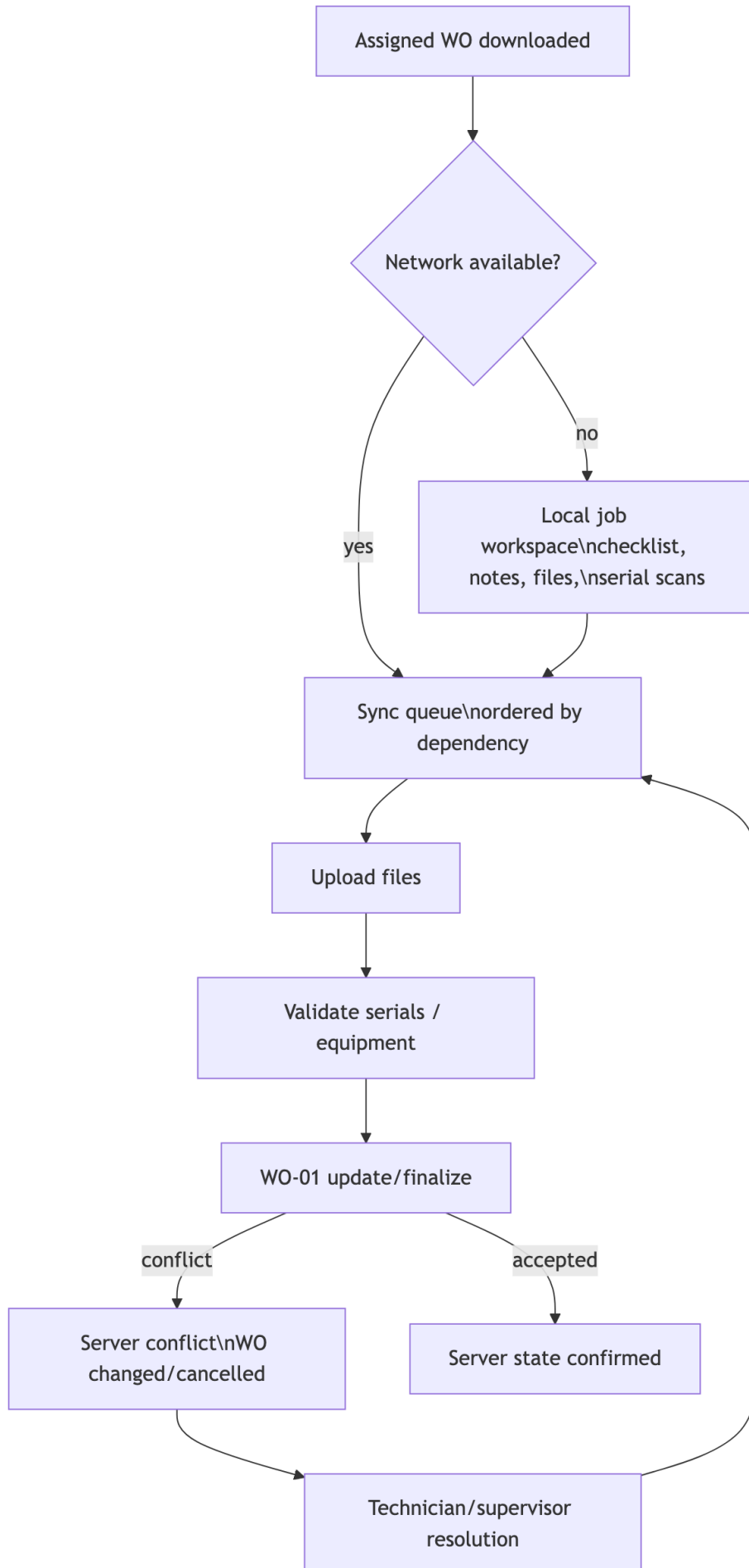
Step	App behavior	Module API link
1. Load jobs	Technician opens assigned/team queue.	GET /api/work-orders?... , EM identity filters.
2. Open WO detail	Show task type, site, contact, checklist, equipment expectations.	GET /api/work-orders/{work_order_id}.
3. Claim/start	Claim team WO or start assigned WO.	POST /api/work-orders/{id}/claim, start/status APIs from WO DD.

Step	App behavior	Module API link
4. Capture evidence	Photos, notes, customer signature, checklist values.	File upload APIs, POST /api/work-orders/{id}/attachments, POST /api/work-orders/{id}/notes.
5. Scan equipment	Validate serials and bind/recover devices.	OSR-INSTANCE-01, POST /api/work-orders/{id}/equipment-bindings.
6. Submit finalization	Final reason/result, checklist, evidence, serials.	WO flow-specific finalization API.
7. Track result	Show completed, pending review, rejected correction, or failed.	WO-01, ICN-01 messages if correction needed.

7.2 Offline Evidence Capture

Illustration:

Figure 4 - Offline sync



Offline mode supports field capture where network is poor. It does not make server state final until sync succeeds.

Rules:

- Cache assigned WO details needed for the day's work.
- Capture photos/signatures/checklist/serial scans locally with local draft ids.
- Upload files before sending WO finalization payload if finalization references file ids.
- Preserve operation ordering: file upload -> OSR serial validation -> WO attachment/equipment binding -> WO finalization.
- If server says WO was cancelled/reassigned/changed, stop sync and show conflict resolution.
- A locally completed job is shown as Pending Sync, not Completed.

7.3 Equipment Swap Or Recovery

Step	App behavior	Module API link
Scan old device	Identify customer/field device.	GET /api/equipment-instances?serial=...
Validate replacement	Confirm replacement device is in technician/team stock and compatible.	OSR-INSTANCE-01, PLM-CFG-06 compatibility where exposed.
Capture swap evidence	Photos, notes, reason, signature.	FOUNDATION_FILE_STORAGE, WO-01.
Submit binding/recovery	Record equipment binding/recovery on WO.	POST /api/work-orders/{id}/equipment-bindings, OSR lifecycle APIs where required.
Finalize	Close WO with swap/RMA outcome.	WO finalization API, OSR-RMA-01 where swap request exists.

7.4 Support Escalation

Step	App behavior	Module API link
Mark cannot resolve	Capture reason and evidence.	WO-01 notes/finalization APIs.
Escalation candidate	WO emits escalation event or exposes allowed action.	WO-01, WorkOrderEscalationCandidate.
Ticket/QCS link	If formal customer case is needed, use ticket lifecycle.	TCK-01; Backoffice may own full handling.
Notify supervisor	Staff task/message created.	ICN-01.

The contractor app should not create a hidden ticket database. If the escalation becomes a customer case, TCK owns it.

8. Screen Groups

8.1 Jobs

Screen/action	Purpose	Module API link	Representative API
Assigned job list	Show WOs assigned to current technician.	WO-01, EM-02.	GET /api/work-orders?assignedTechnicianId={agent_id}&status=...
Team queue	Show claimable WOs for technician's team.	WO-01, EM-02.	GET /api/work-orders?teamId={team_id}&status=PENDING, ASSIGNED.
Urgent/SLA filter	Highlight aging/priority jobs.	WO-01.	Work order query filters by priority/SLA/status.
Refresh/sync	Pull server state and reconcile local drafts.	WO-01, local sync queue.	Job list/detail APIs.

8.2 Work Order Detail

Screen/action	Purpose	Module API link	Representative API
WO summary	Task type, status, phase, priority, due date.	WO-01.	GET /api/work-orders/{work_order_id}.
Customer/site context	Contact, address, HomePass, subscription context.	WO-01 context; ILM/RLM/SUB read APIs if exposed.	WO detail with context panels.

Screen/action	Purpose	Module API link	Representative API
Checklist	Show flow-specific checklist.	W0-01 flow satellite config.	Checklist metadata from WO detail/config API.
Linked ticket/order	Show origin context.	TCK-01, FUL-02, W0-01.	Linked refs from WO detail; owner read APIs if authorized.

8.3 Claim, Start, And Status

Screen/action	Purpose	Module API link	Representative API
Claim WO	Technician claims team-mode job.	W0-01.	POST /api/work-orders/{id}/claim.
Start job	Mark job in progress where DD allows.	W0-01.	WO start/status transition API.
Pause/wait reason	Capture blocked/waiting reason where supported.	W0-01.	WO note/status reason APIs.
Cancel request	Request cancellation where role allows; not direct cancel unless authorized.	W0-01, Backoffice dispatch for approval.	WO cancellation/request APIs.

8.4 Evidence Capture

Screen/action	Purpose	Module API link	Representative API
Photo capture	Capture site/equipment evidence.	FOUNDATION_FILE_STORAGE, W0-01.	File upload APIs, POST /api/work-orders/{id}/attachments.
Customer signature	Capture completion sign-off where required.	FOUNDATION_FILE_STORAGE, W0-01.	File upload + attachment metadata.
Notes	Capture structured/free-text notes.	W0-01.	POST /api/work-orders/{id}/notes.
Checklist answers	Required before finalization.	W0-01.	Checklist/finalization APIs from flow DD.

8.5 Equipment And Stock

Screen/action	Purpose	Module API link	Representative API
Serial scan	Find and validate equipment instance.	OSR-INSTANCE-01.	GET /api/equipment-instances?serial=...
My stock	Show devices held by technician/team/contractor location.	OSR-01, OSR-INSTANCE-01.	GET /api/equipment-instances?operator=WIK&locationId=...
Bind equipment	Attach installed/replaced serial to WO.	W0-01, OSR-INSTANCE-01.	POST /api/work-orders/{id}/equipment-bindings.
Recover equipment	Capture recovered device and reason.	OSR-INSTANCE-01, W0-01.	OSR lifecycle APIs and WO equipment binding/recovery APIs.
Swap/RMA status	Show swap request if linked.	OSR-RMA-01.	GET /api/swap-requests/{swap_request_id}.

8.6 Finalization

Screen/action	Purpose	Module API link	Representative API
Final review	Validate required checklist/files/serials.	Local validation plus WO config.	No authoritative write until submit.
Submit install finalization	Complete installation WO.	W0-01-FLOW-INSTALLATION.	Flow-specific finalize API.
Submit support finalization	Complete support/QCS WO.	W0-01-FLOW-SUPPORT.	Flow-specific finalize API.
Submit shifting phase	Complete disconnect/reconnect phase.	W0-01-FLOW-SHIFTING.	Phase finalization API.
Correction response	Fix rejected/missing evidence.	W0-01, ICN-01.	WO update/attachment APIs and staff task ack.

8.7 Messages

Screen/action	Purpose	Module API link	Representative API
Inbox	Staff messages/tasks.	ICN-01.	GET /api/staff-notifications/inbox?status=DISPATCHED&limit=20.
Acknowledge	Ack received message/task.	ICN-01.	POST /api/staff-notifications/{notification_id}/ack.
Open deep link	Navigate to WO/ticket/order context.	W0-01, TCK-01, FUL-02.	Owner detail APIs.

9. API Integration Details

Required headers:

Header	When	Purpose
Authorization	All protected APIs.	JWT from FOUNDATION_AUTH.
X-Correlation-Id	All important reads and commands.	Trace WO/OSR/file updates.
Idempotency-Key	File commit metadata, equipment binding, finalization submit.	Safe retry from weak networks.
X-Operator-Code	Where endpoint requires explicit operator.	Operator scoping.
Accept-Language	All frontend calls.	Localized errors/messages.

9.1 Evidence Upload And Attachment

```
POST /api/work-orders/wo_01HZ123/attachments
Authorization: Bearer <jwt>
Idempotency-Key: wo_01HZ123_photo_front_001
Content-Type: application/json
```

```
{
  "fileRef": "file_01J_W0_PHOTO_001",
  "attachmentKind": "SITE_PHOTO",
  "capturedAt": "2026-05-28T09:30:00Z",
  "capturedByAgentId": "agt_01HZ_TECH_07",
  "metadata": {
    "geoLat": -1.2831,
    "geoLng": 36.7820,
    "offlineDraftId": "draft_wo_01HZ123_photo_001"
  }
}
```

9.2 Equipment Binding

```
POST /api/work-orders/wo_01HZ123/equipment-bindings
Authorization: Bearer <jwt>
Idempotency-Key: wo_01HZ123_bind_ONT_48575443ABCD1234
Content-Type: application/json
```

```
{
  "equipmentInstanceId": "eqi_01HZZZ",
  "serialNumber": "48575443ABCD1234",
  "bindingRole": "INSTALLED_ONT",
  "action": "BIND_TO_CUSTOMER",
  "capturedByAgentId": "agt_01HZ_TECH_07",
  "evidenceFileRefs": ["file_01J_W0_PHOTO_001"]
}
```

10. Offline And Sync Rules

Allowed offline:

Item	Offline allowed?	Notes
Assigned WO cache	Yes	Cache only jobs assigned/team-visible to the user.
Checklist answers	Yes	Pending until synced.
Notes	Yes	Queue with idempotency key.
Photos/signatures	Yes	Store encrypted local file until upload succeeds.
Serial scan	Yes, as draft	Must validate with OSR before final server submission.

Item	Offline allowed?	Notes
Equipment binding	Limited	Can draft offline; server binding requires online OSR/VO validation.
Finalization	Limited	Can queue as pending sync, but not show completed until WO accepts it.
Claim/start	No, unless online	Avoid two technicians claiming same WO offline.
Cancellation/reassignment	No	Must be online and role-authorized.

Sync order:

1. Refresh WO detail and verify assignment/status still allows updates.
2. Upload files and get file refs.
3. Validate serials/equipment with OSR.
4. Submit WO notes/attachments/equipment bindings.
5. Submit finalization if still valid.
6. Mark local draft synced only after server confirms.

11. Regional And Operator Configuration

Runtime config must control:

Config	Example
Operator/country	WIK, KE.
Currency/locale/timezone	KES, en-KE, Africa/Nairobi.
Brand/theme	Logo, app name, colors.
Enabled WO types	INSTALLATION, SUPPORT, SHIFTING, QCS.
Checklist requirements	Required photos/signatures/serials by WO kind/operator.
Offline policy	Max offline age, allowed offline evidence types.
Map behavior	External map provider/deep link policy.
Feature flags	stockTabEnabled, signatureRequired, qcsEscalationEnabled, offlineFinalizationQueueEnabled.

12. Security And Privacy

Rules:

- Cache only jobs visible to the authenticated user/team.
- Encrypt local job/evidence cache where supported.
- Clear local cache on logout, user switch, or app wipe policy.
- Do not store raw customer ID documents unless the WO explicitly requires it.
- Do not expose customer phone/address to users outside assigned team/role.
- Evidence downloads/uploads must use authorized file APIs.
- A locally cached WO must be revalidated before mutation.
- Do not allow one contractor/franchise to view another contractor's queue unless role/config explicitly allows it.

13. Observability

Track:

Metric/event	Purpose
Job list load failure	Detect WO API/mobile connectivity issues.
Offline draft count and age	Detect stuck field jobs.
Evidence upload failure	Detect file/camera/network issues.

Metric/event	Purpose
Serial validation failure	Detect equipment mismatch/stock problems.
Finalization rejected	Detect checklist/config mismatch.
Claim conflict	Detect assignment race.
Sync duration	Field app reliability.

Logs must include correlation id, work order id, local draft id, and user/role family. Do not log raw customer documents, signatures, or full sensitive notes.

14. E2E Scenarios

Required from DD_QA-01:

Scenario	Expected result
Work order dispatch and finalization	Dispatcher assigns WO, technician executes, evidence uploads, WO finalized.
Equipment swap/RMA	Defective equipment replaced, stock/instance state correct.
Offline evidence sync	Evidence captured offline syncs in order and finalization is accepted once online.
Claim conflict	Two users cannot claim the same WO; losing user sees current server state.
Support escalation	Contractor marks cannot resolve, escalation/ticket path remains owned by WO/TCK.

15. Implementation Notes

Recommended build order:

1. Implement app shell, auth, runtime config, profile/team context.
2. Implement job list and WO detail.
3. Implement local job workspace and sync queue.
4. Implement evidence capture/upload.
5. Implement notes and checklist capture.
6. Implement serial scan and OSR validation.
7. Implement equipment binding/recovery.
8. Implement finalization submit and correction handling.
9. Implement messages/inbox.
10. Add offline sync and E2E tests.

Do not implement finalization before evidence and equipment sync ordering is stable. Otherwise the app can close WOs without the data the downstream modules need.

16. Acceptance Checklist

- [] Every screen/action names its module API owner.
- [] WO-01 is the only owner of WO lifecycle state.
- [] OSR validates equipment before server-side binding/recovery.
- [] Files use FOUNDATION_FILE_STORAGE.
- [] Offline finalization shows Pending Sync, not Completed.
- [] Claim/start requires online server confirmation.
- [] Roles come from FOUNDATION_AUTH; technician/team/franchise identity comes from EM.
- [] Contractor users cannot see other contractor queues unless authorized.
- [] E2E tests cover WO finalization, equipment swap, and offline evidence sync.